

Brief Communication: Proportions of the Ventral Half of the Cerebellar Dentate Nucleus in Humans and Great Apes

Shozo Matano*

Department of Biological Anthropology, Faculty of Human Sciences, Osaka University, Osaka 565-0871, Japan

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ABSTRACT In a previous study about volume comparisons of the cerebellar complex in some hominoid species (1997), progressive development of only the lateral zone group of nuclei was found in the human cerebellar complex. This development was considered to be related not to bipedalism, but to versatile and coordinated finger movement, evolving after bipedalism was established. It was also considered a prerequisite for the evolution of human language. The lateral zone groups of nuclei are represented by the dentate nucleus. Therefore, the present study reports the development of the dentate nucleus in humans in comparison with that in some great

apes. One finding is that the average value of ratios for nucleus size of the ventral half (v) to the dorsal half (d) (v/d) was found to be 2.11 in humans, while it was 1.64 in great apes. This finding shows that the greater part of progressive development of the dentate nucleus in humans is due to the development of its ventral half. Therefore, the fiber connection to the frontal association area from the cerebellar cortex, which is involved in the performance of higher cerebellar functions such as cognitive and language functions, would be mediated by the ventral half of the dentate nucleus. *Am J Phys Anthropol* 114: 163–165, 2001. © 2001 Wiley-Liss, Inc.

In terms of volumetric comparison among various species of primates from Stephan's brain collection, the development of the nucleus in the cerebellar complex and its adaptation to species-specific locomotor behavior have been investigated through allometric comparisons. Seventeen species of prosimians were studied by Matano and Ohta (1999), and 26 species of anthropoids, including great apes and humans, were studied by Matano and Hirasaki (1997).

The latter found that among three longitudinal zones of the cerebellar complex, only the cerebellar hemisphere-lateralis zone, in which the group of nuclei is represented by the dentate nucleus, was prominently developed in humans, compared to that in great apes. This development was considered to be related not to bipedalism, but to versatile and coordinated finger movement, resulting after bipedalism was established. It was also thought to be a prerequisite for the evolution of higher cerebellar functions such as human language and cognition. This higher cerebellar function has been elucidated through neuroimaging and neurobehavioral studies (Leiner et al., 1993; Petersen and Fiez, 1993; Thach, 1996; Carper and Courchesne, 2000).

The present study further compares size development of the dentate nucleus of humans and great apes, using the same samples from Stephan's collection.

MATERIALS AND METHODS

Stephan's collection consists of frontal and sagittal sections of brains fixed in Bouin's fluid, embedded in paraffin, cut at 20 μm serially, and stained with cresyl-violet for cells and by the Heidenhain-Woelcke method for fibers. The details of this collection and raw data of the volume of the dentate nucleus of each species investigated are given by Stephan et al. (1981) and Matano et al. (1985).

In this study, the frontal sections of two human specimens, two *Gorilla* specimens, and one of *Pan* were used from the same samples. The dentate nuclei of the sections were projected onto photographic paper at equal intervals. There were 14 sections from the brain of *Pan*, 15 sections from each brain of *Gorilla*, and 18 sections from each human brain. Using the X-Y digitizer (KW 4620, Graphic Co.), the borders of the nucleus on the photographic paper were traced and its area was estimated in mm^2 . The area of nucleus was divided into two parts, the dorsal half (d) and the ventral half (v), at the dorso-

*Correspondence to: Shozo Matano, M.D., Ph.D., Department of Biological Anthropology, Faculty of Human Sciences, Osaka University, Yamadaoka 1-2, Suita, Osaka 565-0871, Japan.
E-mail: hirasaki@hus.osaka-u.ac.jp

TABLE 1. Size ratios of ventral half (v) to dorsal half (d) (v/d) in the dentate nuclei investigated¹

	Ga	Gb	P	Ha	Hb
Number of sections	15	15	14	18	18
Average v/d from values in each section	1.62	1.63	1.66	2.05	2.16
Range of values (v/d)	1.40–2.02	1.36–2.09	1.31–2.04	1.64–2.43	1.62–2.67

¹ Ga, Gb, *Gorilla*; P, *Pan*; Ha, Hb, *humans*, d, area for dorsal half of dentate nucleus in mm²; v, area for ventral half of nucleus in mm².

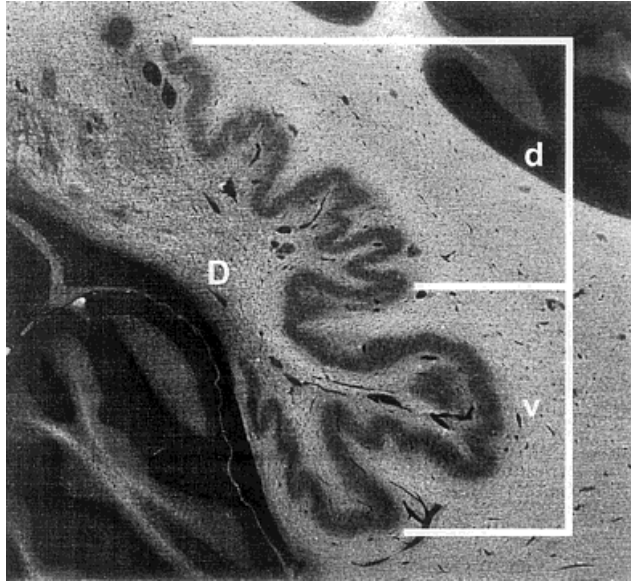


Fig. 1. Definition of ventral half (v) and dorsal half (d) of dentate nucleus (D).

ventral axis of the maximum length of the nucleus in each frontal section (Fig. 1).

The ratio of v to d (v/d) is given in Table 1. Each volume of the dorsal half and the ventral half of the dentate nucleus is given in Table 2. Differences in values for ratios between humans and great apes and humans vs. gorillas were checked by Student's *t*-test. The tests are tentative because the study samples are small.

RESULTS AND DISCUSSION

The dentate nuclei of humans and great apes have been said to have almost the same pattern in shape. No definite difference in the shape of the dentate nuclei of humans and those of great apes was discernible under microscopy. However, size comparisons by allometric analysis in previous studies clearly revealed the prominent development of the human dentate nucleus. For example, the size indices for the medulla oblongata volume of the dentate nucleus of human (2.1) greatly surpass those of great apes (1.4) (Matano and Hirasaki, 1997).

In the present study, values for the ratios of the size of the ventral half of the dentate nucleus to the size of the dorsal half (v/d) were averaged for each case investigated (Table 1). In the two human samples obtained from each of 18 frontal sections, values for the ratio (v/d) were 2.11 on average, whereas they were 1.64 on average for three great ape sam-

TABLE 2. Volume of the dentate nucleus investigated (mm³)

	<i>Gorilla</i>	<i>Pan</i>	Humans
Ventral half	331.33	284.47	791.69
Dorsal half	203.27	171.36	375.21
Total	534.60	455.83	1,166.90

ples. Student's *t*-test indicated significant difference in ventral to dorsal ratio between great apes and humans ($t: 8.59, P < 0.001$). These ratios were also significantly different between humans and gorillas ($t: 8.57, P < 0.001$). Because only two humans and two gorillas (and one chimpanzee) provided the samples, these findings must be interpreted cautiously. Still, the differences suggest real differences in size of ventral relative to dorsal portions of the dental nucleus between humans and great apes.

Some studies indicate existence of fiber connections from the dentate nucleus. Rouiller et al. (1994) reported a cerebello-thalamic cortical projection to the frontal association area in macaque monkeys. Also, Middleton and Strick (1994) showed a projection to the prefrontal area (area 46) from the dentate nucleus by the retrograde tranneuronal transport of HSV₁ (herpes simplex virus type 1) in *Cebus*. They also showed that labelled neurons were confined to the most ventral portion of the middle third of the dentate nucleus. In macaque monkeys, Sasaki (1979) reported the projection of the cerebellar dentate nucleus onto the frontal association cortex by an electrophysiological method.

Higher cerebellar functions involving cognitive and language function have been recognized from neuroimaging and neurobehavioral studies, which were obtained from normal adult cerebella and patients with defective cerebella (Leiner et al., 1993; Petersen and Fiez, 1993; Thach, 1996; Carper and Courchesne, 2000). Functional magnetic resonance imaging showed, in an experiment in learning to use a new tool, that the activity of the lateral cerebellar cortex was confined to an area near to the posterior superior fissure (Imamizu et al., 2000).

The evidence in humans thus indicates the presence of a projection from the lateral cerebellar cortex to the frontal association area via the thalamus. Also, the output for performing such higher cerebellar functions as has evolved in humans is mediated by the dentate nucleus. Its ventral half, which appears to be much more developed in humans than in great apes, may play an important role in fiber connection. The preliminary findings described here need to confirmed by using larger samples.

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